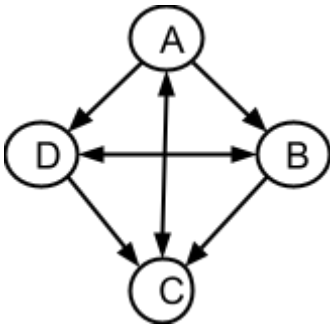
American Computer Science League

2022-2023 ● Contest 4: Short Problems ● Junior Division

1. Graph Theory

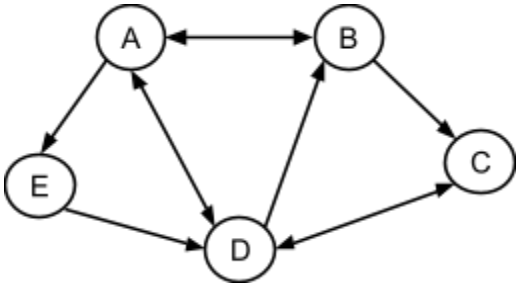
How many paths of length 3 are there in the following directed graph?



A. 20
B. 22
C. 24
D. 26
E. 28

2. Graph Theory

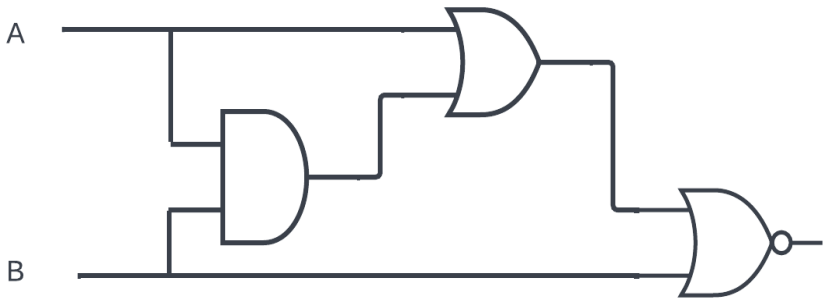
How many cycles are there from A in the directed graph below?



A. 2
B. 3
C. 4
D. 5
E. 6

3. Digital Electronics

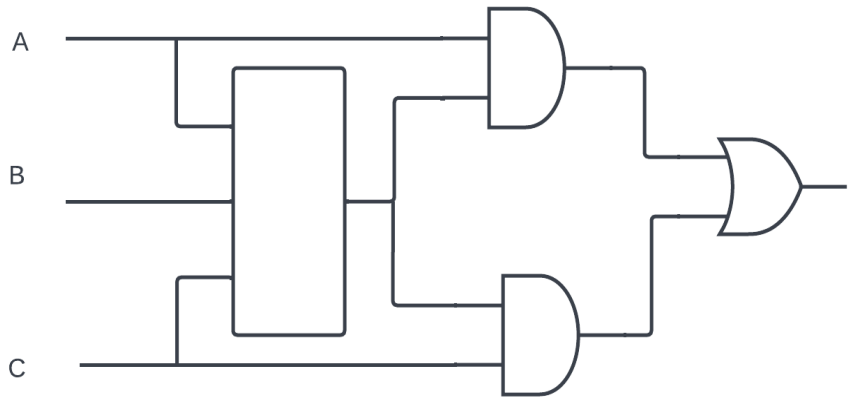
Simplify the Boolean expression represented by the following digital circuit:



A. $\overline{A}$
B. $\overline{B}$
C. $\overline{A} + \overline{B}$
D. $\overline{A}\,\overline{B}$
E. $\overline{A}\,B$

4. Digital Electronics

How many ordered triples make the following digital circuit TRUE if a new gate (rectangle) is defined to be TRUE if exactly 2 of the 3 inputs are TRUE?



A. 0
B. 1
C. 2
D. 3
E. 4

5. What Does This Program Do? (Strings)

Define the functions val(x) to get the numeric value of a numeric string and str(x) to make a numeric value into a string. For example, val("9") = 9 and str(9) = "9". If s1 = "637456" and s2 = "345167", what will be the output for the following program?

```
input s1, s2
c = 0
ans = ""
for x = len(s1)-1 to 0 step -1
     t = val(s1[x]) + val(s2[x]) + c
     c = 0
     if t >= 8 then
          t = t - 8
          c = 1
     end if
     ans = str(t) + ans
next x
if c == 1 then
     ans = "1" + ans
end if
output ans
```

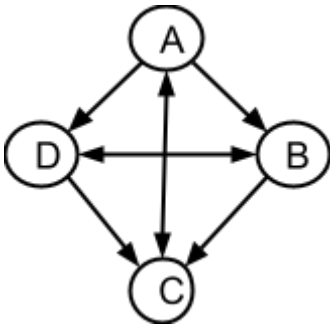
A. 0174535
B. 1204645
C. 1274635
D. 1174535
E. 0204645

American Computer Science League

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1. 图论

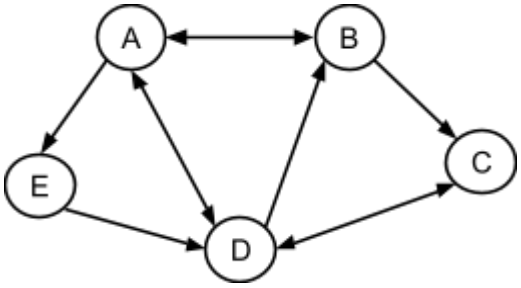
在下述有向图中有多少条长度为 3 的路径？



A. 20
B. 22
C. 24
D. 26
E. 28

2. 图论

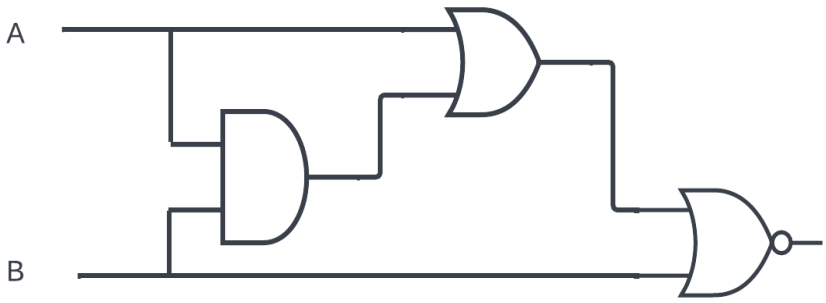
在下述有向图中，请问从 A 出发有多少个环？



A. 2
B. 3
C. 4
D. 5
E. 6

3. 数字电路

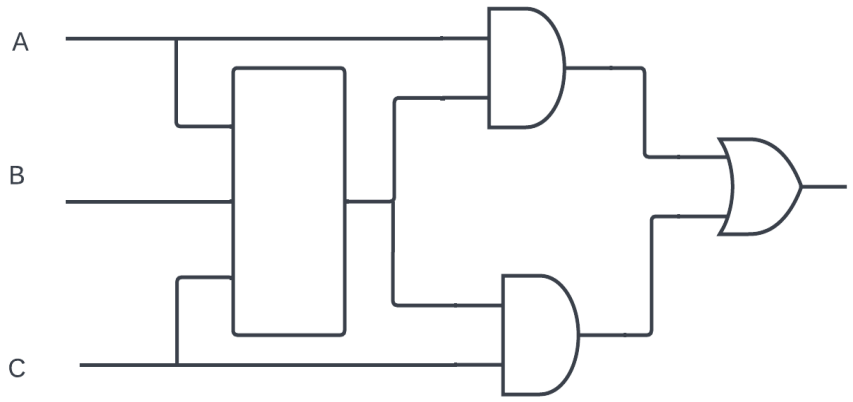
化简下述数字电路表示的布尔表达式。



A. $\overline{A}$
B. $\overline{B}$
C. $\overline{A} + \overline{B}$
D. $\overline{A}\,\overline{B}$
E. $\overline{A}\,B$

4. 数字电路

如果一个新的门（矩形）的 3 个输入中**恰**好有 2 个为**真**，则这个门被定义为**真**，那么有多少**个**有序三元组可以使以下数字电路为**真**？



A. 0
B. 1
C. 2
D. 3
E. 4

5. 这个程序的作用是什么？（字符串）

定义函数 val（x）用来获取**一个**数**字**字符串的数值，定义函数 str（x）将数值转换为字符串。例如，val（“9”）= 9，str（9）= “9”。如果 s1 = “637456”，s2 = “345167”，以下程序**会**输出什么？

```
input s1, s2
c = 0
ans = ""
for x = len(s1)-1 to 0 step -1
     t = val(s1[x]) + val(s2[x]) + c
     c = 0
     if t >= 8 then
          t = t - 8
          c = 1
     end if
     ans = str(t) + ans
next x
if c == 1 then
     ans = "1" + ans
end if
output ans
```

A. 0174535
B. 1204645
C. 1274635
D. 1174535
E. 0204645